

1. Provide summary statistics (mean, median, minimum, maximum, standard deviation) for a dataset (age, income etc.) with numeric variables grouped by one of the qualitative (categorical) variables. For example, if your categorical variable is age groups and quantitative variable is income, then provide summary statistics of income grouped by the age groups. Create a list that contains a numeric value for each response to the categorical variable.

```
import pandas as pd
import numpy as np
from sklearn import preprocessing

df = pd.read_csv("Mall_Customers.csv")

df
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	
1	2	Male	21	15	
2	3	Female	20	16	
3	4	Female	23	16	
4	5	Female	31	17	
...	...	...	...	...	...
195	196	Female	35	120	
196	197	Female	45	126	
197	198	Male	32	126	
198	199	Male	32	137	
199	200	Male	30	137	

```
[200 rows x 5 columns]

df.columns

Index(['CustomerID', 'Gender', 'Age', 'Annual Income (k$)',
      'Spending Score (1-100)'],
      dtype='object')

df.describe()
```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

```
df.isnull().sum()
```

```
CustomerID      0
Gender          0
Age            0
Annual Income (k$) 0
Spending Score (1-100) 0
dtype: int64
```

```
# Compute summary statistics for Annual Income grouped by Gender
```

```
summary_stats = df.groupby('Gender')['Annual Income (k$)'].describe()
[['mean', '50%', 'min', 'max', 'std']]
summary_stats.rename(columns={'50%': 'median'}, inplace=True)
print(summary_stats)
```

	mean	median	min	max	std
Gender					
Female	59.250000	60.0	16.0	126.0	26.011952
Male	62.227273	62.5	15.0	137.0	26.638373

```
# Create a dictionary with numeric values for each gender category
```

```
income_list = df.groupby('Gender')['Annual Income  
(k$)'].apply(list).to_dict()  
print(income_list)
```

```
{ 'Female': [16, 16, 17, 17, 18, 18, 19, 19, 20, 20, 21, 23, 25, 28,
28, 29, 29, 30, 33, 33, 34, 34, 37, 37, 38, 39, 39, 39, 40, 40, 40,
40, 42, 43, 43, 44, 46, 47, 47, 48, 48, 48, 49, 50, 50, 54, 54, 54,
54, 54, 57, 57, 58, 58, 59, 60, 60, 60, 60, 62, 62, 62, 63, 63, 64,
65, 65, 65, 65, 67, 67, 67, 69, 70, 70, 72, 72, 73, 73, 74, 75, 76,
76, 77, 78, 78, 78, 78, 78, 78, 78, 79, 79, 81, 85, 86, 87, 88, 88,
```

```
97, 97, 98, 99, 101, 103, 103, 103, 103, 113, 120, 120, 126], 'Male':
[15, 15, 19, 19, 20, 20, 21, 23, 24, 24, 25, 28, 28, 30, 33, 33, 38,
39, 42, 43, 43, 44, 46, 46, 46, 48, 48, 48, 49, 54, 54, 54, 54, 54,
54, 54, 59, 60, 60, 61, 61, 62, 62, 62, 63, 63, 63, 63, 64, 67, 69,
71, 71, 71, 71, 71, 71, 73, 73, 74, 75, 77, 77, 77, 78, 78, 78, 78,
78, 81, 85, 86, 87, 87, 87, 87, 87, 88, 88, 93, 93, 98, 99, 101, 113,
126, 137, 137]]}
```

2. Write a Python program to display some basic statistical details like percentile, mean, standard deviation etc. of the species of 'Iris-setosa', 'Iris-versicolor' and 'Iris-versicolor' of iris.csv dataset.

```
df = pd.read_csv("iris.csv")
print("Dataset Preview:\n", df.head())

Dataset Preview:
   sepal_length  sepal_width  petal_length  petal_width  species
0           5.1           3.5           1.4           0.2   setosa
1           4.9           3.0           1.4           0.2   setosa
2           4.7           3.2           1.3           0.2   setosa
3           4.6           3.1           1.5           0.2   setosa
4           5.0           3.6           1.4           0.2   setosa

# Group by Species and Provide Summary Statistics for Numeric Columns
summary = df.groupby('species').agg(['mean', 'median', 'min', 'max',
'std'])
print("\nSummary Statistics by Grouping Categorical Variable:\n")
print(summary)
```

Summary Statistics by Grouping Categorical Variable:

\	sepal_length					sepal_width		
	mean	median	min	max	std	mean	median	
min								
species								
setosa	5.006	5.0	4.3	5.8	0.352490	3.428	3.4	
2.3								
versicolor	5.936	5.9	4.9	7.0	0.516171	2.770	2.8	
2.0								
virginica	6.588	6.5	4.9	7.9	0.635880	2.974	3.0	
2.2								
petal_width \	petal_length							
	max	std	mean	median	min	max	std	
mean								

```
species
setosa      4.4  0.379064      1.462  1.50  1.0  1.9  0.173664
0.246
versicolor  3.4  0.313798      4.260  4.35  3.0  5.1  0.469911
1.326
virginica   3.8  0.322497      5.552  5.55  4.5  6.9  0.551895
2.026
```

```
      median  min  max      std
species
setosa      0.2  0.1  0.6  0.105386
versicolor  1.3  1.0  1.8  0.197753
virginica   2.0  1.4  2.5  0.274650
```

```
# Display Statistical Details for Each Species
print("\nStatistical Details for Iris-setosa:")
print(df[df['species'] == 'Iris-setosa'].describe())

print("\nStatistical Details for Iris-versicolor:")
print(df[df['species'] == 'Iris-versicolor'].describe())

print("\nStatistical Details for Iris-virginica:")
print(df[df['species'] == 'Iris-virginica'].describe())
```

Statistical Details for Iris-setosa:

	sepal_length	sepal_width	petal_length	petal_width
count	0.0	0.0	0.0	0.0
mean	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN

Statistical Details for Iris-versicolor:

	sepal_length	sepal_width	petal_length	petal_width
count	0.0	0.0	0.0	0.0
mean	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN

Statistical Details for Iris-virginica:

	sepal_length	sepal_width	petal_length	petal_width
count	0.0	0.0	0.0	0.0
mean	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN